

agent = model + harness; the harness is everything below

THE MODEL: reasoning only (Claude session, Mode B) — no memory between calls, no ground truth

1 run the loop

burst-walkthrough protocol: RUN → PRESENT → GATE → LITERATURE → DISTILL

dispatcher: roster, gates, order at task intake

prototype chains: fit, merge, products, temporal, assembly

queue manager + typed workflows (A17)

the repository layer

2 execute tools

spectral engine: 24-model menu × every usable bin (campaign run), validity + rail columns

Bayesian blocks (3ML native) + significance merge

Stage-1 approval GUI + ingest: detectors, background, source — decision human or AI, stamped

figures, temporal chain, SED family, per-burst report

3 manage context

27 guide files; 10 step skills indexed by the ledger; lessons with IDs; memory files + notes

skill-reader: the binding checklist at every step open

standing contracts in the PI's quoted words (S-, R-items)

isolation: each verifier and reader runs in its own fresh context

4 persist state

state board derived from DISK evidence, never from memory

identity stamps, receipts, sidecars, commit pin

object schemas (10) + validation test; action trace with model id

first-class actions: intent → preflight → commit → receipt

5 enforce boundaries

hooks: no-ship (sha), dispatch plan, raw-data guard

producer ≠ verifier ≠ approver; fresh-context verifiers

fit-table auditor, tie rule, count coordinates, edge stamps

loop caps, spend ledger, post-tool audit, pre-commit, sandbox

PI gate: every step

identity-stamped into the records above; never fabricated

hosted coding product: runtime loop, tool calls, compaction, permissions

code: deterministic, tested

model judgement: a role in fresh context

documents, records: read or written, not run

human

dashed: PROPOSED

solid = on disk at the census (2026-09-02); each box maps to rows of Table T1; hatched frames = the two harness layers; grey headers = the five jobs